

EXPERIMENT – 2

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Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1: INSTRUCTION PAGE

Chunking Analysis of the Instruction Page

Chunking is a cognitive technique that organizes information into smaller, meaningful units. The Instruction Page of the Memory Recall Task effectively applies chunking as follows:

1. Clear and Sequential Instructions

Instructions are divided into numbered steps (1–6).

Each step focuses on one action only, reducing cognitive load.

step-by-step flow helps users understand the task easily.

2. Logical Grouping of Information

Observation Phase (Steps 1–2): Explains what users will see.

Memorization Phase (Step 3): Encourages focusing on remembering items.

Recall Phase (Steps 4–5): Explains how recall will happen.

Performance Metric (Step 6): Emphasizes accuracy over speed.

3. Visual Hierarchy and Design

Large bold title “MEMORY RECALL TASK” clearly defines the activity.

Proper spacing and bullet points improve readability.

A highlighted START button guides the user to proceed.

Cartoon visuals (Angry Birds theme) add engagement without distraction.

4. Time Constraint Awareness

The instruction “You will have 5 seconds” prepares users mentally.

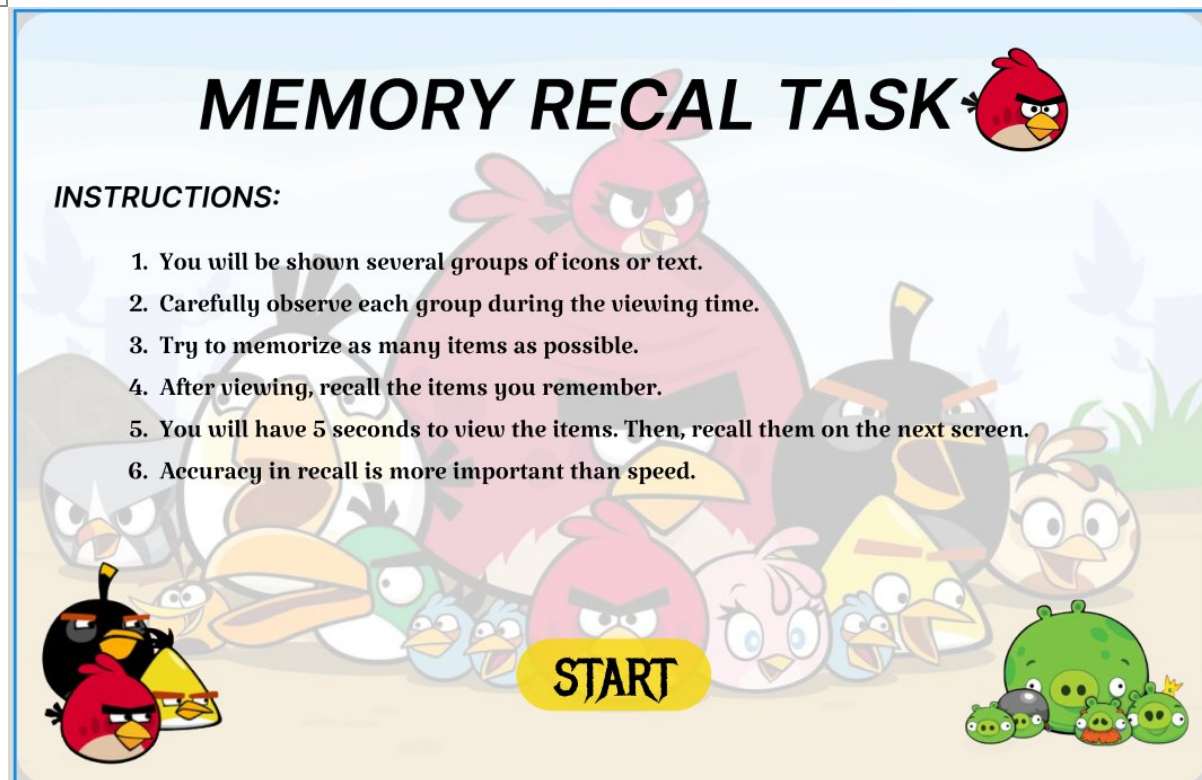
This improves attention and focus during the task.

5. Simplicity and Clarity

Short, simple sentences are used.

Active voice makes instructions easy to follow.

Result: Users clearly understand the task with minimal effort.



FRAME 2: CHUNKING PHASE (MEMORIZATION SCREEN)

Analysis of the Chunking Phase

This screen represents the visual encoding phase, where users observe and memorize icons.

1. Purpose of the Screen

Users view multiple icons within a limited time.

Encourages users to mentally group similar icons using chunking.

2. Key UI Elements

Countdown Timer (Top Left – 0.05)

Shows remaining time to observe icons.

Creates urgency and improves focus.

Progress Bar

Visually represents time passing.

Helps users manage attention.

Grid of Icons

Icons are displayed in a structured grid layout.

Users can apply chunking strategies such as:

Grouping by color

Grouping by character type

Recognizing repeated patterns

Game Theme (Angry Birds Characters)

Familiar characters improve recognition.

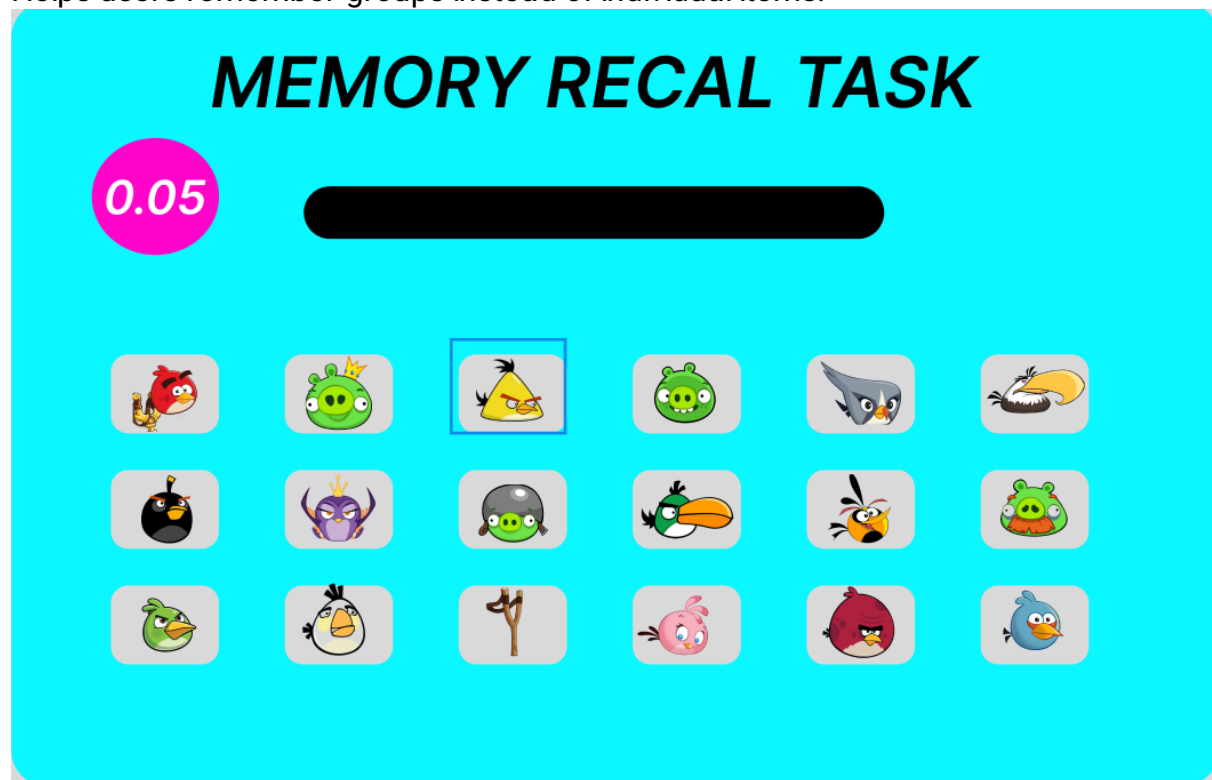
Reduces cognitive strain and increases engagement.

3. Cognitive Benefits

Improves short-term memory retention.

Reduces information overload.

Helps users remember groups instead of individual items.



FRAME 3: RECALL PHASE (SELECTION SCREEN)

Analysis of the Recall Phase

This screen tests how effectively users remembered items from the Chunking Phase.

1. Purpose

Users must identify previously seen icons.

Measures accuracy of memory recall.

2. UI Components

Title & Instruction

“SELECT THE ITEMS YOU REMEMBER” gives clear direction.

Icon Selection Grid

Includes:

Icons shown earlier

Distractor icons (new items)

Users select remembered icons using radio buttons.

Submit Button

Confirms user choices.

Leads to score evaluation.

3. How It Works

Users recall icons from memory.

Select matching icons.

Distractors test false memory.

Submit final answers.

4. UX & Cognitive Benefits

Tests memory accuracy.

Encourages focused decision-making.
Gamified design keeps users engaged.



FRAME 4: RESULT PAGE (SCORE & FEEDBACK)

Analysis of the Result Screen

This screen provides feedback on memory performance.

1. Purpose

Displays recall accuracy.

Allows user to choose next action.

2. UI Elements

Score Display

Shows score clearly (e.g., 5/5).

Wooden board design makes results visually appealing.

Action Buttons

CONTINUE: Move to next level

RESTART: Retry the task

EXIT: End the activity

Gamified Theme

Friendly visuals reduce stress.

Encourages repeated attempts.

3. Cognitive Benefits

Immediate feedback improves learning.

User control increases motivation.

Reinforces the effectiveness of chunking.

MEMORY RECAL TASK

Your Recall Score:



PHOTOTYPE LINK:

<https://www.figma.com/proto/Odoa3BxhA8Nkd1cMMagzi/Untitled?node-id=1-2&p=f&t=q5FWUfmmABY4szJN-0&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A2>